

YAONAN JIN

jinyaonan@ust.hk

EDUCATION

Columbia University PhD in Computer Science, advised by Xi Chen and Rocco Servedio Thesis: <i>Bayesian Auction Design and Approximation</i>	2019.09 – 2023.04
Hong Kong University of Science and Technology MPhil in Operations Research, advised by Qi Qi Thesis: <i>Tight Approximation Ratio of Anonymous Pricing</i>	2017.09 – 2018.12
Shanghai Jiao Tong University BEng in Computer Science	2013.09 – 2017.06

CAREER HISTORY

Hong Kong University of Science and Technology Assistant Professor of Computer Science	2026.06 – present
Huawei's Taylor Lab (Shanghai) Researcher (Huawei's Top Minds Scheme)	2023.05 – 2026.05

RESEARCH INTERESTS

Theoretical Computer Science
Economics and Computation
Online Algorithms
Combinatorial Optimization

INVITED ARTICLES

- [S1] Yaonan Jin and Pinyan Lu
Settling the Efficiency of First Price Auction [\[PDF\]](#)
ACM SIGecom Exchanges, 20(2): 69–74, 2022
- [S2] Yaonan Jin, Pinyan Lu, Qi Qi, Zhihao Tang, and Tao Xiao
Tight Revenue Gaps among Simple and Optimal Mechanisms [\[PDF\]](#)
ACM SIGecom Exchanges, 17(2): 54–61, 2019

JOURNAL PAPERS

* In Theoretical Computer Science, authors are listed alphabetically.
Names of student authors mentored are underlined.

- [J1] Yaonan Jin and Pinyan Lu
Benchmark-Tight Approximation Ratio of Simple Mechanism for a Unit-Demand Buyer
To appear in SIAM Journal on Computing
- [J2] Yaonan Jin and Pinyan Lu
First Price Auction is $1 - 1/e^2$ Efficient [\[arXiv\]](#)
Journal of the ACM, 70(5): 36:1-36:86

- [J3] Jarosław Błasiok, Peter Ivanov, Yaonan Jin, Chin Ho Lee, Rocco Servedio, and Emanuele Viola
Fourier Growth of Structured $\mathbb{F}_2$ -Polynomials and Applications [\[arXiv\]](#) [\[ECCC\]](#)
(**Invited Paper**) To appear in Theory of Computing
- [J4] Yaonan Jin, Shunhua Jiang, Pinyan Lu, and Hengjie Zhang
Tight Revenue Gaps among Multi-Unit Mechanisms [\[arXiv\]](#)
SIAM Journal on Computing, 51(5): 1535–1579, 2022
- [J5] Nick Gravin, Yaonan Jin, Pinyan Lu, and Chenhao Zhang
Optimal Budget-Feasible Mechanisms for Additive Valuations [\[arXiv\]](#)
ACM Transactions on Economics and Computation, 8(4): 1–15, 2020
- [J6] Yaonan Jin, Pinyan Lu, Zhihao Tang, and Tao Xiao
Tight Revenue Gaps among Simple Mechanisms [\[arXiv\]](#)
SIAM Journal on Computing, 49(5): 927–958, 2020

CONFERENCE PAPERS

- [C1] Xihan Deng, Yaonan Jin, Wenqian Wang, and Yuhao Zhang
The Price of Anarchy for Selfish Load Balancing
67th IEEE Symposium on Foundations of Computer Science (FOCS 2026)
- [C2] Sayan Bhattacharya, Martín Costa, Ermiya Farokhnejad, Shaofeng H.-C. Jiang, Yaonan Jin, and Jianing Lou
Fully Dynamic Euclidean k -Means [\[arXiv\]](#)
67th IEEE Symposium on Foundations of Computer Science (FOCS 2026)
- [C3] Yaonan Jin and Yingkai Li
Anonymous Pricing in Large Markets [\[arXiv\]](#)
27th ACM Conference on Economics and Computation (EC 2026)
- [C4] Houshuang Chen, Yaonan Jin, Pinyan Lu, and Chihao Zhang
Tight Regret Bounds for Fixed-Price Bilateral Trade [\[arXiv\]](#)
53rd International Colloquium on Automata, Languages, and Programming (ICALP 2026)
- [C5] Houshuang Chen, Yaonan Jin, Pinyan Lu, and Chihao Zhang
The Query Complexity of Uniform Pricing [\[arXiv\]](#)
35th ACM Web Conference (WWW 2026)
- [C6] Shaofeng H.-C. Jiang, Yaonan Jin, Jianing Lou, and Pinyan Lu
Local Search for Clustering in Almost-linear Time [\[arXiv\]](#)
37th ACM-SIAM Symposium on Discrete Algorithms (SODA 2026)
- [C7] Yiding Feng and Yaonan Jin
Beyond Regularity: Simple versus Optimal Mechanisms, Revisited [\[arXiv\]](#)
66th IEEE Symposium on Foundations of Computer Science (FOCS 2025)
- [C8] Yaonan Jin and Pinyan Lu
Benchmark-Tight Approximation Ratio of Simple Mechanism for a Unit-Demand Buyer
65th IEEE Symposium on Foundations of Computer Science (FOCS 2024)
- [C9] Xi Chen, Yaonan Jin, Tim Randolph, and Rocco Servedio
Subset Sum in Time $2^{n/2}/\text{poly}(n)$ [\[arXiv\]](#)
27th International Workshop on Randomization and Computation (RANDOM 2023)
- [C10] Yaonan Jin, Pinyan Lu, and Tao Xiao
Learning Reserve Prices in Second Price Auctions [\[arXiv\]](#)
14th Innovations in Theoretical Computer Science Conference (ITCS 2023)

- [C11] Yaonan Jin, Daogao Liu, and Zhao Song
Super-Resolution and Robust Sparse Continuous Fourier Transform in Any Constant Dimension: Nearly Linear Time and Sample Complexity [\[arXiv\]](#)
 34th ACM-SIAM Symposium on Discrete Algorithms (SODA 2023)
- [C12] Yaonan Jin and Pinyan Lu
The Price of Stability for First Price Auction [\[arXiv\]](#)
 34th ACM-SIAM Symposium on Discrete Algorithms (SODA 2023)
- [C13] Yaonan Jin and Pinyan Lu
First Price Auction is $1 - 1/e^2$ Efficient [\[arXiv\]](#)
 63rd IEEE Symposium on Foundations of Computer Science (FOCS 2022)
- [C14] Xi Chen, Yaonan Jin, Tim Randolph, and Rocco Servedio
Average-Case Subset Balancing Problems [\[arXiv\]](#)
 33rd ACM-SIAM Symposium on Discrete Algorithms (SODA 2022)
- [C15] Jarosław Błasiok, Peter Ivanov, Yaonan Jin, Chin Ho Lee, Rocco Servedio, and Emanuele Viola
Fourier Growth of Structured $\mathbb{F}_2$ -Polynomials and Applications [\[arXiv\]](#) [\[ECCC\]](#)
 25th International Workshop on Randomization and Computation (RANDOM 2021)
- [C16] Yaonan Jin, Shunhua Jiang, Pinyan Lu, and Hengjie Zhang
Tight Revenue Gaps among Multi-Unit Mechanisms [\[arXiv\]](#)
 22nd ACM Conference on Economics and Computation (EC 2021)
- [C17] Yaonan Jin, Weian Li, and Qi Qi
On the Approximability of Simple Mechanisms for MHR Distributions [\[PDF\]](#)
 15th Conference on Web and Internet Economics (WINE 2019)
- [C18] Nick Gravin, Yaonan Jin, Pinyan Lu, and Chenhao Zhang
Optimal Budget-Feasible Mechanisms for Additive Valuations [\[arXiv\]](#)
 20th ACM Conference on Economics and Computation (EC 2019)
- [C19] Yaonan Jin, Pinyan Lu, Qi Qi, Zhihao Tang, and Tao Xiao
Tight Approximation Ratio of Anonymous Pricing [\[arXiv\]](#)
 51st ACM Symposium on Theory of Computing (STOC 2019)
- [C20] Yaonan Jin, Pinyan Lu, Zhihao Tang, and Tao Xiao
Tight Revenue Gaps among Simple Mechanisms [\[arXiv\]](#)
 30th ACM-SIAM Symposium on Discrete Algorithms (SODA 2019)

MANUSCRIPTS

- [M1] Xihan Deng, Yaonan Jin, Wenqian Wang, and Yuhao Zhang
The Price of Anarchy for Selfish Load Balancing under Correlated Priors
- [M2] Shaofeng H.-C. Jiang, Yaonan Jin, Jianing Lou, and Weicheng Wang
Round-efficient Fully-scalable MPC algorithms for k-Means [\[arXiv\]](#)
- [M3] Yaonan Jin
Tight Regret Bounds for Bilateral Trade under Semi Feedback [\[arXiv\]](#)
- [M4] Yaonan Jin, Yingkai Li, Yining Wang, and Yuan Zhou
On Asymptotically Tight Tail Bounds for Sums of Geometric and Exponential Random Variables [\[arXiv\]](#)

GRANTS

- Huawei, Computing System Theory and Technology
Title: *Welfare and Revenue Guarantees of Practical Auctions*
Period: 2024.01 – 2025.12, Amount: 3,526,000 CNY

TALKS GIVEN

- *Anonymous Pricing in Large Markets*
 - Shanghai Jiao Tong University, Theory Seminar, June 2026
- *Tight Regret Bounds for Fixed-Price Bilateral Trade*
 - National University of Singapore, Theory Seminar, January 2026
 - Tsinghua University, Yau Mathematical Sciences Center, November 2025
 - Renmin University of China, Gaoling School of Artificial Intelligence, November 2025
 - SUFE ITCS Workshop, June 2025
 - Huawei's Taylor Lab, Theory Seminar, May 2025
 - CUHK, Theory Seminar, April 2025
 - HKUST, IEDA+CS, Theory Seminar, April 2025
 - HKU, Theory Seminar, April 2025
- *Local Search for Clustering in Almost-linear Time*
 - Nanyang Technological University, Theory Seminar, January 2026
 - New York University Shanghai, Theory Seminar, December 2025
 - Institute of Software (CAS), Theory Seminar, November 2025
 - **Invited Talk**, HCP 2025, August 2025
- *Beyond Regularity: Simple versus Optimal Mechanisms, Revisited*
 - FOCS 2025, December 2025
 - Peking University, Center on Frontiers of Computing Studies, November 2025
 - New York University Shanghai, Theory Seminar, October 2025
 - City University of Hong Kong, Theory Seminar, April 2025
 - Nanyang Technological University, Theory Seminar, November 2024
 - National University of Singapore, Department of Economics, November 2024
 - HKUST-Guangzhou, Information Hub, November 2024
 - Shanghai Jiao Tong University, Theory Seminar, November 2024
 - Huawei's Taylor Lab, Theory Seminar, November 2024
- *Benchmark-Tight Approximation Ratio of Simple Mechanism for a Unit-Demand Buyer*
 - National University of Singapore, Theory Seminar, November 2024
 - FOCS 2024, October 2024
 - HKUST, IEDA, Theory Seminar, August 2024
 - **Invited Talk**, IJTCS 2024, July 2024

- SUFE ITCS Workshop, June 2024
- Renmin University of China, Gaoling School of Artificial Intelligence, December 2023
- Institute of Computing Technology (CAS), Theory Seminar, December 2023
- Tsinghua University, Yau Mathematical Sciences Center, December 2023
- Peking University, Center on Frontiers of Computing Studies, December 2023
- University of Science and Technology of China, Theory Seminar, December 2023
- Huawei’s Taylor Lab, Theory Seminar, November 2023
- *Bayesian Auction Design and Approximation*
 - Columbia PhD Thesis Defense, April 2023
 - Columbia PhD Thesis Proposal, January 2023
- *The Price of Stability for First Price Auction*
 - SODA 2023, January 2023
- *Learning Reserve Prices in Second Price Auctions*
 - ITCS 2023, January 2023
- *First Price Auction is $1 - 1/e^2$ Efficient*
 - **Invited Tutorial**, COCOON 2024, August 2024
 - Tsinghua University, Institute for Interdisciplinary Information Sciences, December 2023
 - University of Science and Technology of China, Theory Seminar, December 2023
 - Shanghai Jiao Tong University, Theory Seminar, December 2023
 - Fudan University, School of Data Science, November 2023
 - **Invited Talk**, EC 2023 “Highlights beyond EC” Sessions, July 2023
 - Institute of Computing Technology (CAS), Theory Seminar, December 2022
 - TCS+ Talk, November 2022
 - FOCS 2022, October 2022
 - Harvard EconCS Seminar, October 2022
 - UPenn Theory Seminar, October 2022
 - Princeton Theory Seminar, October 2022
 - Columbia Theory Seminar, September 2022
- *Bayesian Multi-Unit Revenue Maximization*
 - Columbia PhD Candidacy Exam, May 2022
- *Tight Revenue Gaps among Multi-Unit Mechanisms*
 - EC 2021, July 2021
- *On the Approximability of Simple Mechanisms for MHR Distributions*
 - WINE 2019, December 2019
- *Optimal Budget-Feasible Mechanisms for Additive Valuations*

- EC 2019, June 2019
- SUFE Theory Seminar, April 2019
- *Tight Approximation Ratio of Anonymous Pricing*
 - China Computer Federation, TCS PhD Forum, June 2021
 - STOC 2019, June 2019
 - SUFE Theory Seminar, April 2019
 - HKUST, IEDA, Reading Group, December 2018
- *Tight Revenue Gaps among Simple Mechanisms*
 - Northwestern Theory Seminar, March 2019
 - SODA 2019, January 2019
 - HKU Theory Seminar, December 2018
 - SUFE Theory Seminar, December 2017
 - HKUST, IEDA, Reading Group, October 2017

PROFESSIONAL SERVICES

Program Committee

- WINE 2026, ICALP 2026, EC 2026, WWW 2026, WINE 2025, EC 2025

Journal Reviewer

- SIAM Journal on Computing, Algorithmica, Artificial Intelligence, ACM Transactions on Economics, Operations Research Letters

Conference Reviewer

- FOCS, STOC, SODA, EC, ITCS, ISAAC, WWW, WINE, SAGT

Service to Huawei's Taylor Lab

- Panelist: Huawei Collaborative Research Project (with University Faculty), 6 panels
- Theory seminar organizer, 2023.05 – 2026.05

SELECTED HONORS

Featured in Huawei 2024 Annual Report (p. 68) [PDF] for Research Highlights on [C8]	2025
Invited Survey [S1] in ACM SIGecom Exchanges for Research Highlights on [C13]	2022
Invited Survey [S2] in ACM SIGecom Exchanges for Research Highlights on [C19][C20]	2019
National Scholarship (1st Prize), Shanghai Jiao Tong University	2014

TEACHING EXPERIENCES

Columbia University

- COMS 4246: Introduction to Computational Complexity Fall 2020
Graduate course in complexity theory (TA for Xi Chen)
- COMS 4995: Incentives in Computer Science Spring 2020
Graduate course in algorithmic economics (TA for Tim Roughgarden)

Shanghai University of Finance and Economics

- 1665/101389: Discrete Mathematics Spring 2019
Undergraduate course in discrete mathematics (TA for Nick Gravin)
- 0008/213583: Mathematical Tools and Efficient Algorithms Summer 2018
Graduate course in graph and algebraic algorithms (TA for Richard Peng)

Hong Kong University of Science and Technology

- IEDA 3300: Industrial Data Systems Fall 2018
Undergraduate course in database (TA for Qi Qi)
- IEDA 3300: Industrial Data Systems Spring 2018
Undergraduate course in database (TA for Qi Qi)

Shanghai Jiao Tong University

- CS 358: Data Structures Fall 2016
Undergraduate course in data structures (TA for Yong Yu)
- CS 358: Data Structures Fall 2015
Undergraduate course in data structures (TA for Yong Yu)